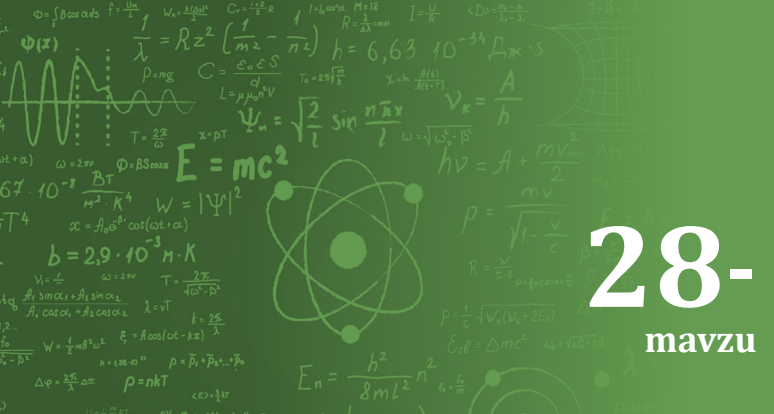




28-

mavzu

MASALALAR YECHISH

1 8 kg massali jism tik yuqoriga qarab 5 m balandlikka tekis ko'tarildi. Jismni ko'tarishda qanday ish bajarilgan?

Berilgan:	Formula	Hisoblash
$m = 8 \text{ kg}$ $h = 5 \text{ m}$ $g = 9,81 \text{ N/kg}$	$A = m g h$ $[A] = \text{kg} \cdot \frac{\text{N}}{\text{kg}} \cdot \text{m} = \text{N} \cdot \text{m} = \text{J}$	$A = 8 \cdot 9,81 \cdot 5 = 392,4 \text{ J}$ Javob: $A = 392,4 \text{ J}$
Topish kerak: $A = ?$		

2 Hajmi $0,5 \text{ m}^3$ bo'lgan marmar plitani 6 m balandlikka tik ravishda tekis ko'tarish uchun qanday ish bajariladi?

Berilgan:	Formula	Hisoblash
$V = 0,5 \text{ m}^3$ $h = 6 \text{ m}$ $\rho = 2700 \text{ kg/m}^3$ $g = 9,81 \text{ N/kg}$	$m = \rho V$ $A = m g h$ $[m] = \frac{\text{kg}}{\text{m}^3} \cdot \text{m}^3 = \text{kg}$ $[A] = \text{kg} \cdot \frac{\text{N}}{\text{kg}} \cdot \text{m} = \text{N} \cdot \text{m} = \text{J}$	$m = 2700 \cdot 0,5 \text{ kg} = 1350 \text{ kg}$ $A = 1350 \cdot 9,81 \cdot 6 \text{ J} = 79461 \text{ J}$ Javob: $A = 79461 \text{ J}$
Topish kerak: $A = ?$		

3 Ot aravani 500 N kuch bilan, 5 m/s tezlikda tekis tortib ketmoqda. Ot 0,5 soatda qanday ish bajargan?

Berilgan:	Formula	Hisoblash
$F = 500 \text{ N}$ $v = 5 \text{ m/s}$ $t = 0,5 \text{ h} = 1800 \text{ s}$	$s = v \cdot t$ $A = F \cdot s$ $[s] = \frac{\text{m}}{\text{s}} \cdot \text{s} = \text{m}$ $[A] = \text{N} \cdot \text{m} = \text{J}$	$s = 5 \cdot 1800 \text{ m} = 9000 \text{ m}$ $A = 500 \cdot 9000 \text{ J} = 4500000 \text{ J}$ Javob: $A = 4,5 \text{ MJ}$
Topish kerak: $A = ?$		



20-mashq

1 20 kg massali jism 5 m balandlikka tekis ko'tarildi. Jismni ko'tarishda qanday ish bajarilgan?

2 Daraxtning shoxida o'tirgan, massasi 5 kg bo'lgan burgutning potensial energiyasi 0,4 kJ bo'lsa, u qanday balandlikda turibdi?

3 Suvning yerga urilishda oqim tezligi 20 m/s bo'lgan sharshara qanday balandlikdan tushmoqda?

4 Jism 5 N kuch ta'sirida 4 m balandlikka tekis ko'tarildi. Shu jarayonda qancha ish bajarildi?

5 Massasi 1,5 tonna bo'lgan avtomobil tezligini 36 km/h dan 72 km/h gacha o'zgartirdi. Avtomobilning kinetik energiyasi qanchaga o'zgargan?

6 Yerdan 10000 metr balandlikda 900 km/h tezlik bilan uchib ketayotgan samolyotning to'liq energiyasini toping. (Samolyot massasi 30 000 kg).

7 10 metr balandlikda uchib ketayotgan qaldirg'ochning to'la energiyasi 30 J teng. Qaldirg'ochning kinetik energiyasini aniqlang. (Qaldirg'ochning massasi 200 gramm).

